5 - Advanced Graphics

Perception

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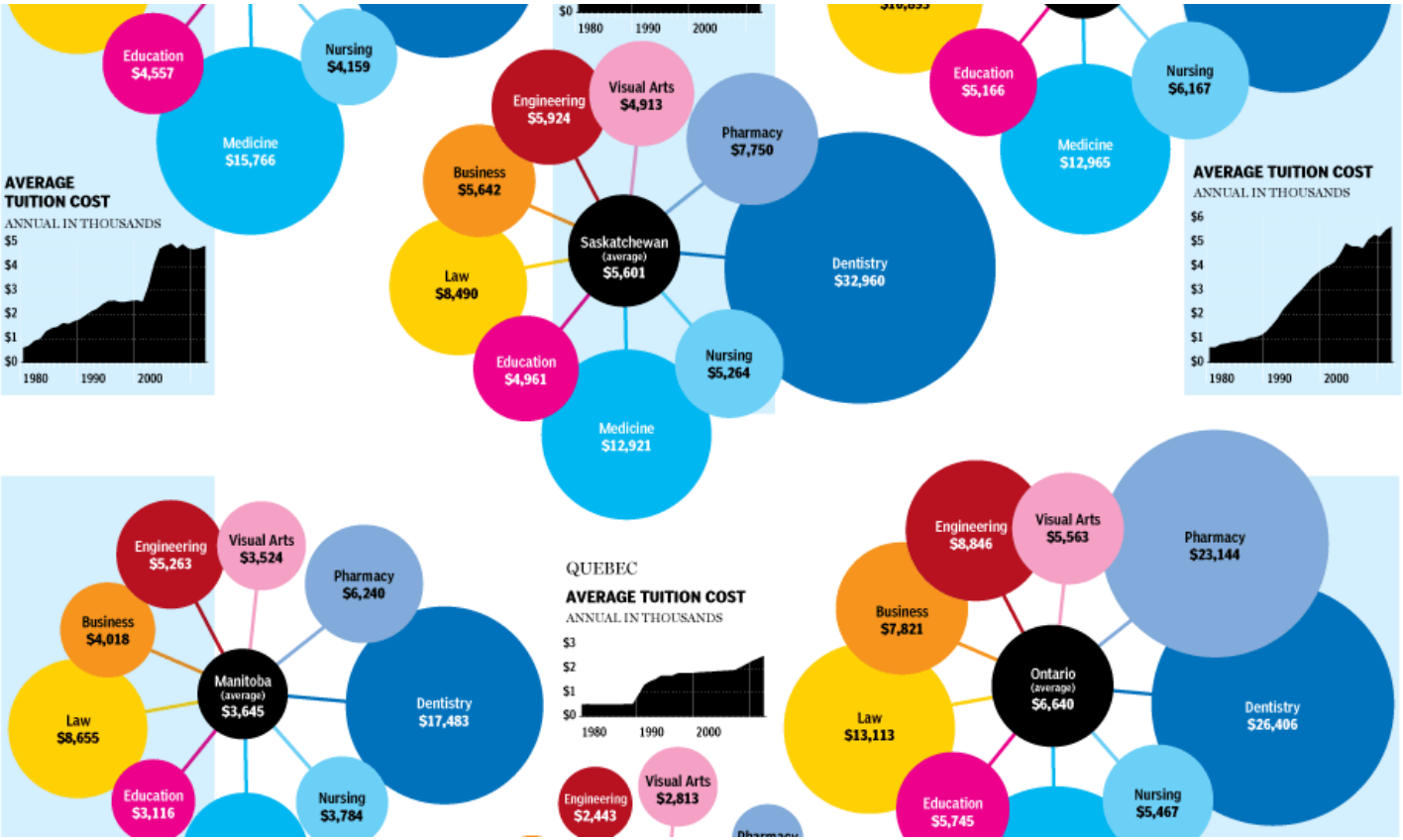
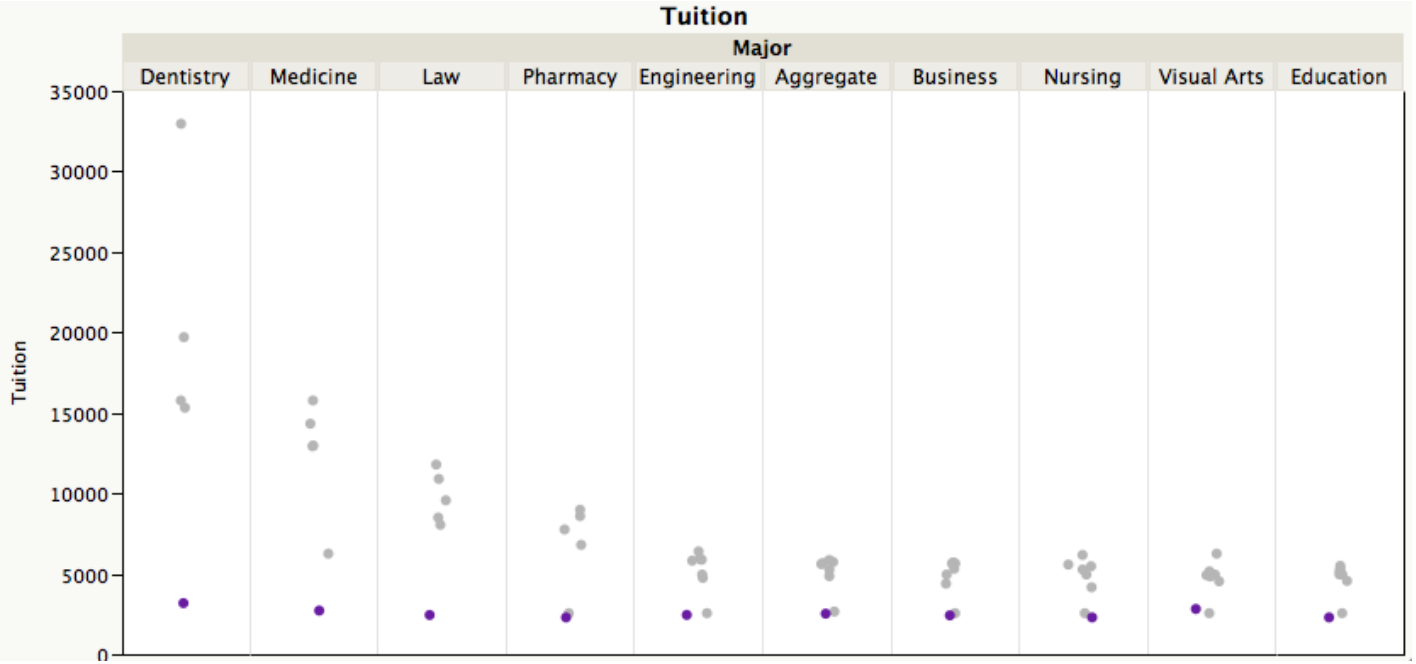
March 12, 2014

Cost of an Education



Motivation

- Why are some plots easier to read?



- http://junkcharts.typepad.com/junk_charts/2012/05/spring-flowers-and-striking-hours.html

Good Graphics

Graphics consist of

- Structure (boxplot, scatterplot, etc.)
- Aesthetics: features such as color, shape, and size that map other characteristics to structural features

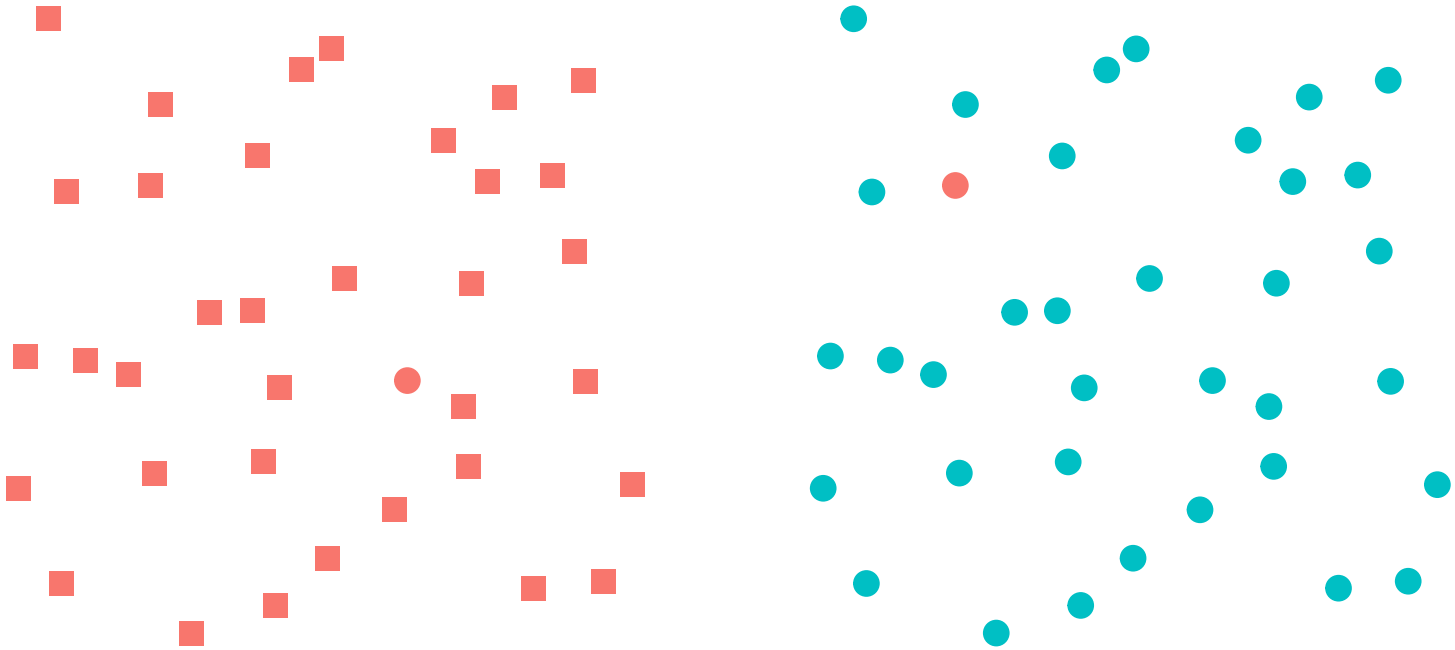
Both the structure and aesthetics should help viewers interpret the information.

Outline

- Cognitive aspects of perception and aesthetic choices
- Visual ordering mechanisms and color choices
- Faceting graphs to show additional variables

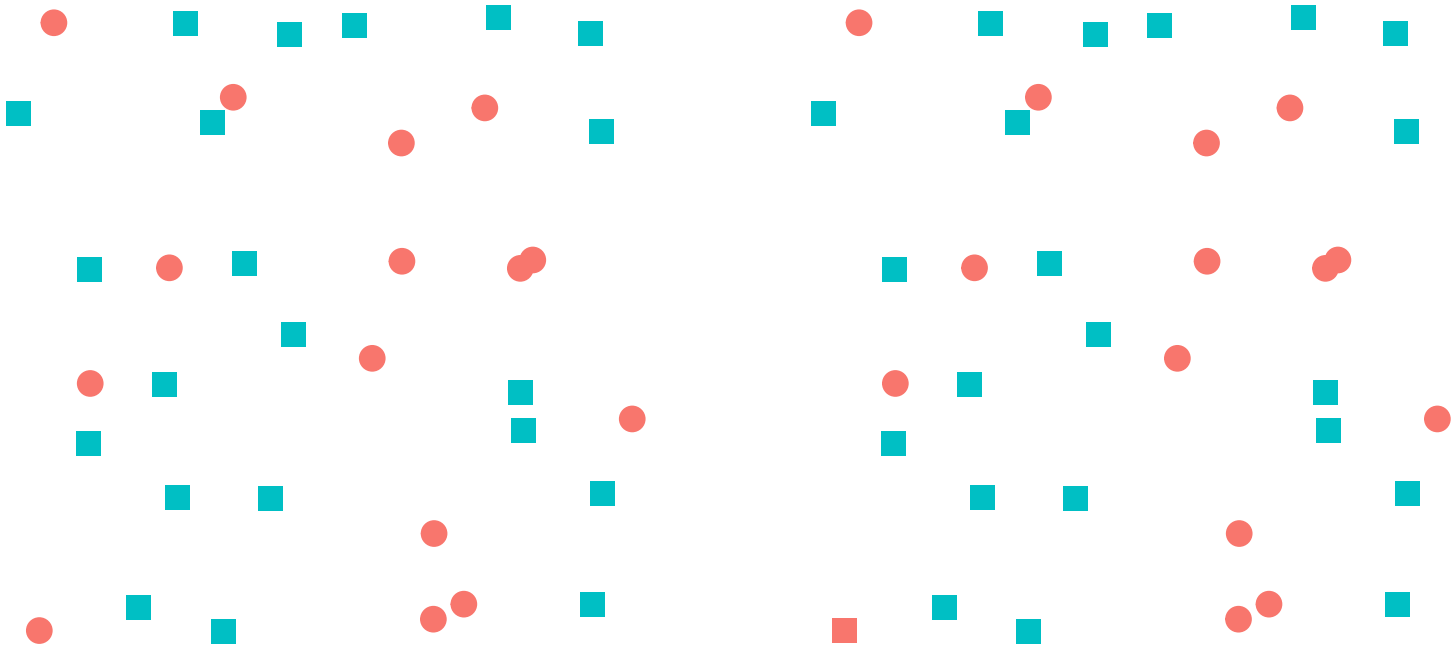
Pre-Attentive Features

- Things that “jump out" in less than 250 ms
- Color, form, movement, spatial localization



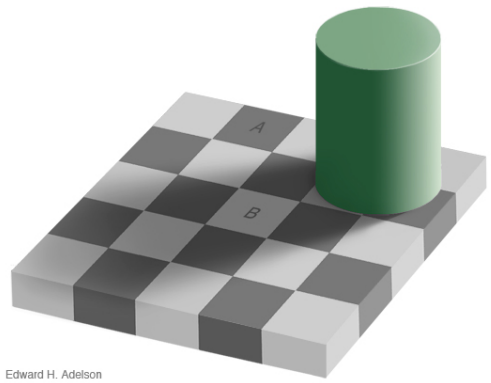
Hierarchy of Features

- Color is stronger than shape
- Combinations of pre-attentive features are usually not pre-attentive due to *interference*



Color

- Hue: shade of color (red, orange, yellow...)
- Intensity: amount of color
- Both color and hue are pre-attentive. Bigger contrast corresponds to faster detection.



Color is context-sensitive: the exact same hue and intensity in one situation may appear to be a different color in a different context. A and B are the same intensity and hue, but appear to be different.

Aesthetics in ggplot2

Scales

Scales control the mapping between data and aesthetics. Read more

- scale_alpha — Alpha scale for continuous variable
- scale_continuous — Continuous position scale
- scale_datetime — Position scale, date time
- scale_gradient — Smooth gradient between two colours
- scale_gradientn — Smooth gradient between n colours
- scale_hue — Qualitative colour scale with evenly spaced hues
- scale_linetype — Scale for line patterns
- scale_shape — Scale for shapes, aka glyphs
- scale_brewer — Sequential, diverging and qualitative colour scales from colorbrewer.org
- scale_date — Position scale, date
- scale_discrete — Discrete position scale
- scale_gradient2 — Smooth gradient between three colours (high, low and midpoints)
- scale_grey — Sequential grey colour scale
- scale_identity — Use values without scaling
- scale_manual — Create your own discrete scale
- scale_size — Size scale for continuous variable

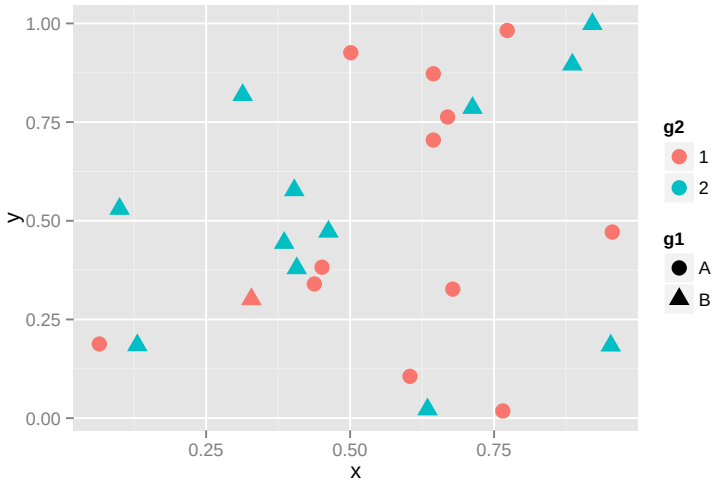
Main parameters: alpha, shape, color, size

Your Turn

Find ways to improve the following graphic:

```
frame <- data.frame(x=runif(25), y=runif(25),
        g1=rep(c("A","B"), c(12,13)),
        g2=rep(c("1", "2"), c(13,12)))
qplot(x,y, shape=g1, colour=g2, data=frame, size=I(4))
```

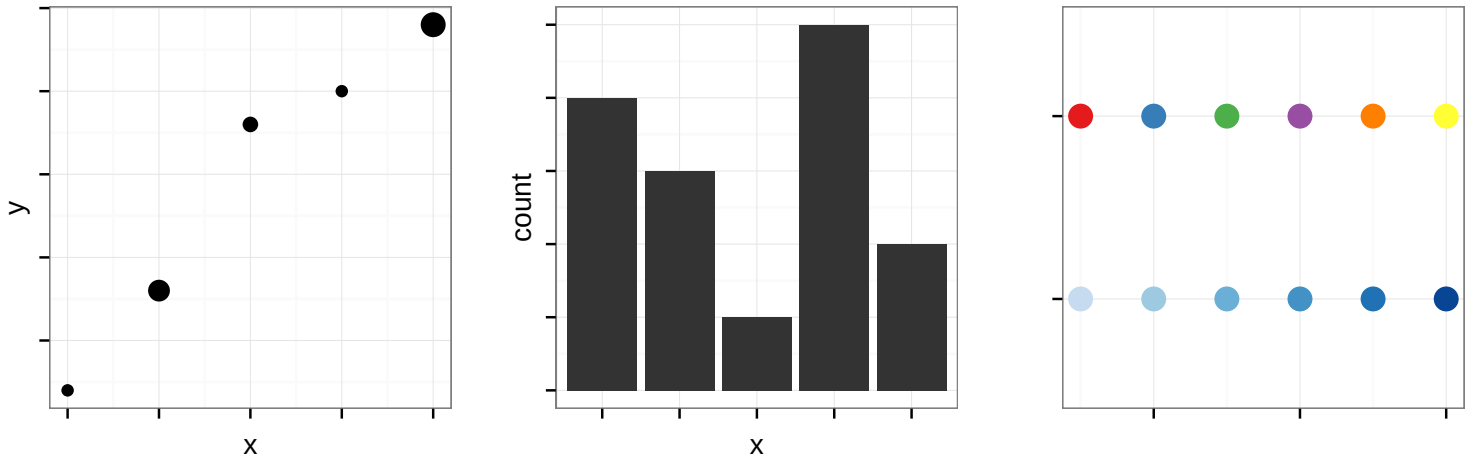
- Make sure the “oddball" stands out while keeping the information on the groups
- Hint: `interaction` combines factor variables



Ordering Variables

Which is bigger?

- Position: higher is bigger (y), items to the right are bigger (x)
- Size, Area
- Color: not always ordered. More contrast = bigger.
- Shape: Unordered.

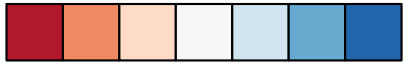


Using Color

- Qualitative schemes: no more than 7 colors

- Quantitative schemes:
  - use color gradient with only one hue for positive values

  - use color gradient with two hues for positive and negative values. Gradient should go through a light, neutral color (white)

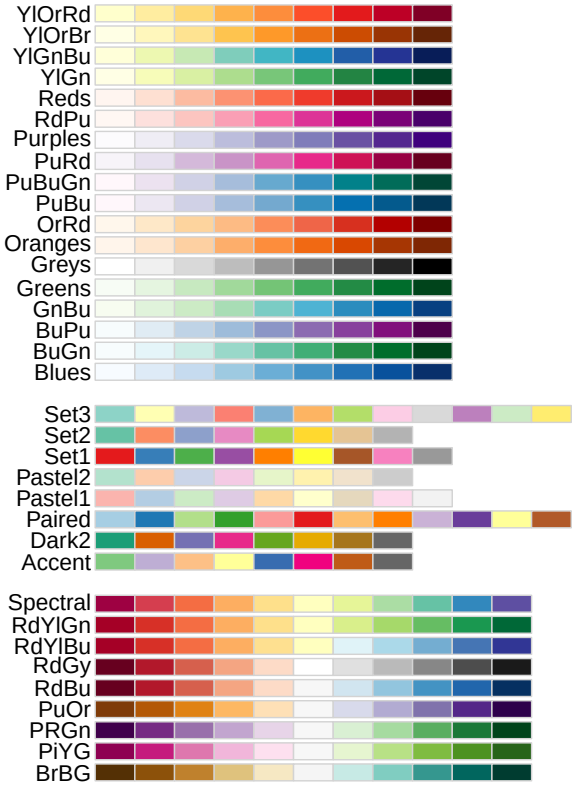


- Small objects or thin lines need more contrast than larger areas

RColorBrewer

R package based on Cynthia Brewer's color schemes (ColorBrewer2.org)

```
install.packages("RColorBrewer")
library(RColorBrewer)
help(package=RColorBrewer)
display.brewer.all()
```



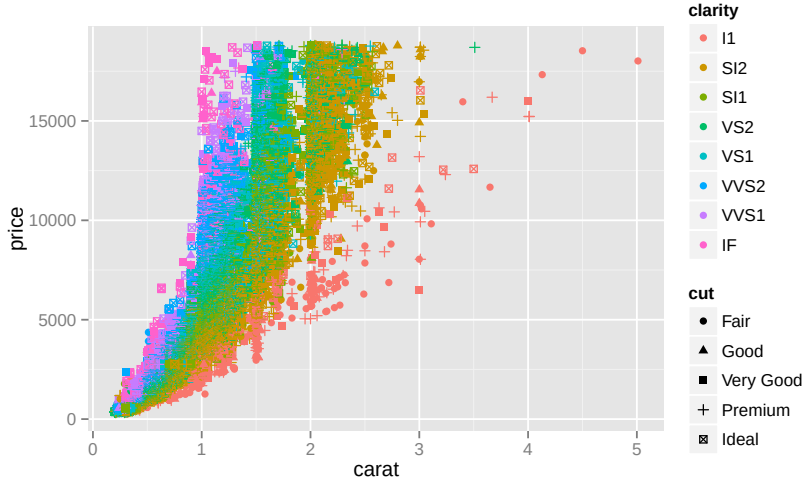
Color in ggplot2

- factor variable:
  `scale_colour_discrete`
  `scale_colour_brewer(palette=...)`
- continuous variable:
  `scale_colour_gradient` (define low, high values)
  `scale_colour_gradient2` (define low, mid, and high values)
- equivalents for fill: `scale_fill_...`

Your Turn

- In the diamonds data, clarity and cut are ordinal, while price and carat are continuous
- Find a graphic that gives an overview of these four variables while respecting their types
- Hint: Start with

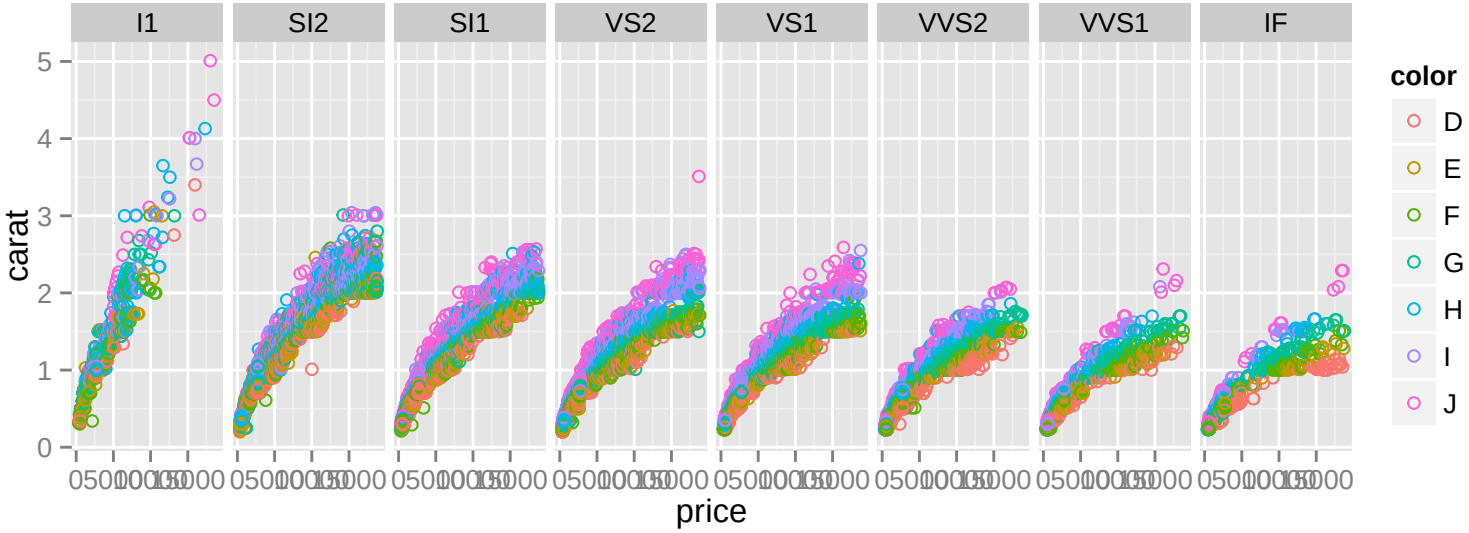
```
data(diamonds)
qplot(carat, price, shape=cut, colour=clarity,
      data=diamonds)
```



Facetting

- A way to extract subsets of data and place them side-by-side in graphics
- Syntax: `facets = row ~ col` Use `.` if there is no variable for either row or column (i.e. `facets = . ~ col`)

```
qplot(price, carat, data=diamonds, color=color,
      facets = . ~ clarity)
```



Your Turn

- The file `errors.csv` contains data for comparing errors from four estimators in a total of 6 situations: two different sample sizes, three different distributions
- Give a visual comparison of the estimators in different simulation situations. The primary interest is the different estimators, the effect of distributions is secondary
- Hint: use facetting!